

Eugene Moshchyn

Dr. Shangqing Zhao

CS 5173, Section 1

February 4, 2026

Homework 1

Problem 1 Explain the three major security objectives: availability, integrity and confidentiality. Which one is more important than the others in the following scenarios?

(a) Suppose your job is to maintain the security of the patient information/data system for the OU-HSC

(b) What if you are managing Norman's emergency alert system?

Reading: https://en.wikipedia.org/wiki/2018_Hawaii_false_missile_alert

Solution

Solution...

Problem 2 Explain DDoS Attacks and their impact on the Internet. What are the possible countermeasures against such attacks?

Solution

Solution...

Problem 3 Does the frequency analysis attack also work on Vigenère cipher? Why?

Solution

Solution...

Problem 4 If a system is perfectly secure, can we say the system cannot be cracked in any way? Why?

Solution

Solution...

Problem 5 Prove one-time pad is perfectly secure.

Solution

Solution...

Problem 6 Suppose Alice designs a one-time pad variant: the plaintext is no longer XOR, but bite-wise multiply the one-time key that is generated uniformly at random. For example:

Plaintext:	0 0 1 1 0 1
Key:	1 0 1 1 1 0
Ciphertext:	0 0 1 1 0 0

Is Alice's design good?

Hint: you should explain if $P(M | C) \stackrel{?}{=} P(M)$ by using mathematical proof. Only textual explanation without the inclusion of mathematical formulas will result in an incomplete analysis and, therefore, may lead to a deduction in grading.

Solution

Solution...

Problem 7 Is using the same key twice in one-time pad secure (i.e., Encryption: $M \oplus K \oplus K = C$; Decryption: $C \oplus K \oplus K = M$)? Why? Please explain in detail. (Hint: you should explain if $P(M | C) \stackrel{?}{=} P(M)$ by using mathematical proof. Only textual explanation without the inclusion of mathematical formulas will result in an incomplete analysis and, therefore, may lead to a deduction in grading.)

Solution

Solution...